

Magnetic Fields

Prepared by:

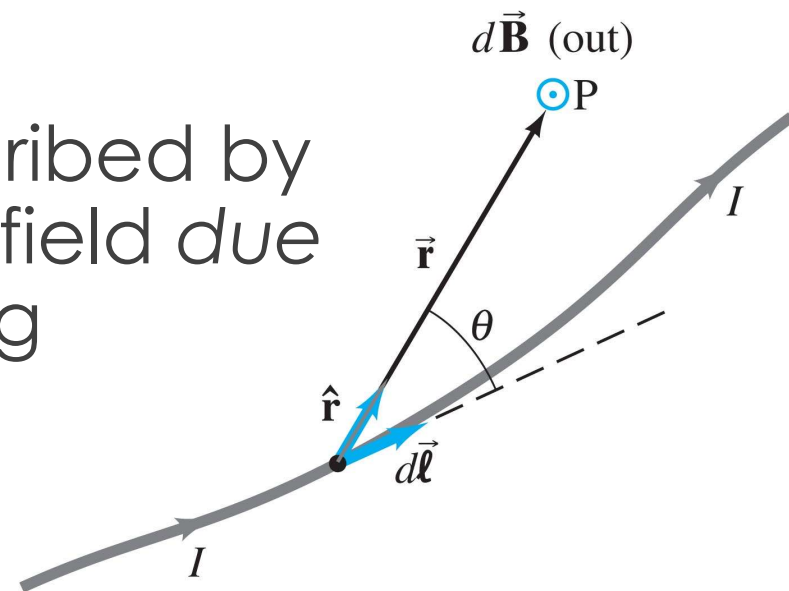
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Magnetic Fields

- ▶ The origin of the magnetic field is moving charges.
- ▶ The magnetic field due to various current distributions can be calculated.
- ▶ Magnetic effects in matter can be explained on the basis of atomic magnetic moments.

Biot-Savart Law

- ▶ Biot and Savart conducted experiments on the force exerted by an electric current on a nearby magnet.
- ▶ The magnetic field described by the Biot-Savart Law is the field *due* to a given current carrying conductor.



Biot-Savart Law

- ▶ The vector $d\mathbf{B}$ is perpendicular both to $d\mathbf{l}$ (which points in the direction of the current) and to the unit vector along $\mathbf{r}$ directed from $d\mathbf{l}$ to P .
- The magnitude dB of the field $d\mathbf{B}$ is inversely proportional to r^2 , where r is the distance from $d\mathbf{l}$ to P .
- The magnitude dB of the field $d\mathbf{B}$ is proportional to the current and to the magnitude ds of the length element $d\mathbf{l}$.
- The magnitude dB of the field $d\mathbf{B}$ is proportional to $\sin\theta$, where θ is the angle between the vectors $d\mathbf{l}$ and $\mathbf{r}$.

Biot-Savart Law – Equation

► The observations are summarized in the mathematical equation called the **Biot-Savart law**:

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{Id\mathbf{l} \times \hat{\mathbf{r}}}{r^2}$$

► The constant μ_0 is called the **permeability of free space**.

► $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m} / \text{A}$

Total Magnetic Field

- ▶ To find the total field, sum up the contributions from all the current elements I

$$\mathbf{B} = \frac{\mu_0}{4\pi} \int \frac{I d\mathbf{l} \times \hat{\mathbf{r}}}{r^2}$$

- ▶ The integral is over the entire current distribution.
- ▶ The law is also valid for a current consisting of charges flowing through space.

Magnetic Field Compared to Electric Field

► Distance

- The magnitude of the magnetic field varies as the inverse square of the distance from the source.
- The electric field due to a point charge also varies as the inverse square of the distance from the charge.

► Direction

- The electric field created by a point charge is radial in direction.
- The magnetic field created by a current element is perpendicular to both the length element $d\mathbf{s}$ and the unit vector $\hat{\mathbf{r}}$

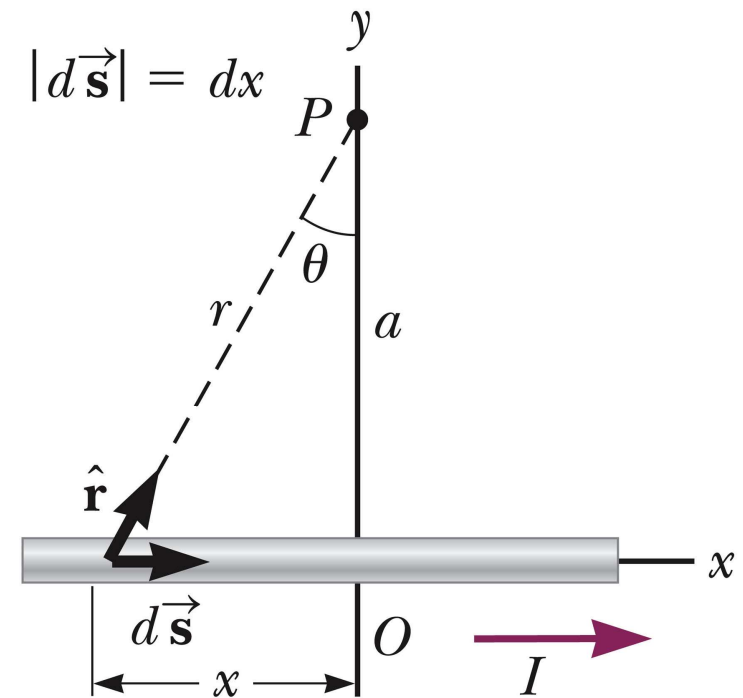
Magnetic Field for a Long, Straight Conductor

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I ds x \hat{\mathbf{r}}}{r^2}$$

$$(ds x \hat{\mathbf{r}}) = |ds x \hat{\mathbf{r}}| \hat{\mathbf{k}} = (dx \sin \theta) \hat{\mathbf{k}}$$

$$d\mathbf{B} = \frac{\mu_0 I}{4\pi} \frac{(dx \sin \theta)}{r^2} \hat{\mathbf{k}}$$

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \int \frac{(dx \sin \theta)}{r^2} \hat{\mathbf{k}} = \frac{\mu_0 I}{4\pi} \left(\int \frac{(dx \sin \theta)}{r^2} \right) \hat{\mathbf{k}}$$



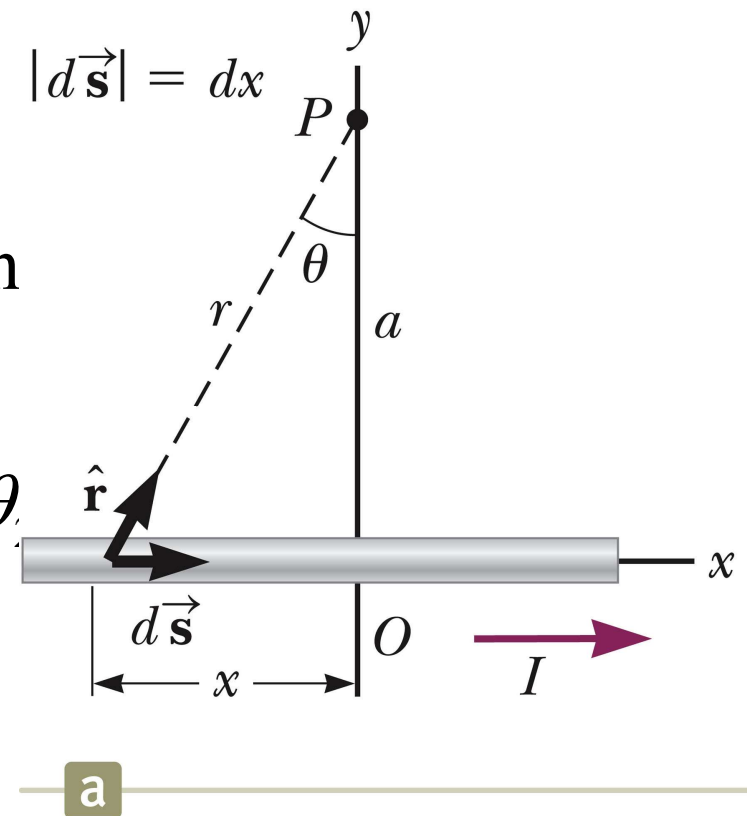
a

Magnetic Field for a Long, Straight Conductor

$$\int \frac{dx \sin \theta}{r^2} = \int_{\theta_1}^{\theta_2} \frac{a \csc^2 \theta d\theta \sin \theta}{a^2 \csc^2 \theta} = \frac{1}{a} \int_{\theta_1}^{\theta_2} d\theta \sin \theta$$

$$\int \frac{dx \sin \theta}{r^2} = \frac{1}{a} [-\cos \theta]_{\theta_1}^{\theta_2} = \frac{1}{a} [\cos \theta_1 - \cos \theta_2]$$

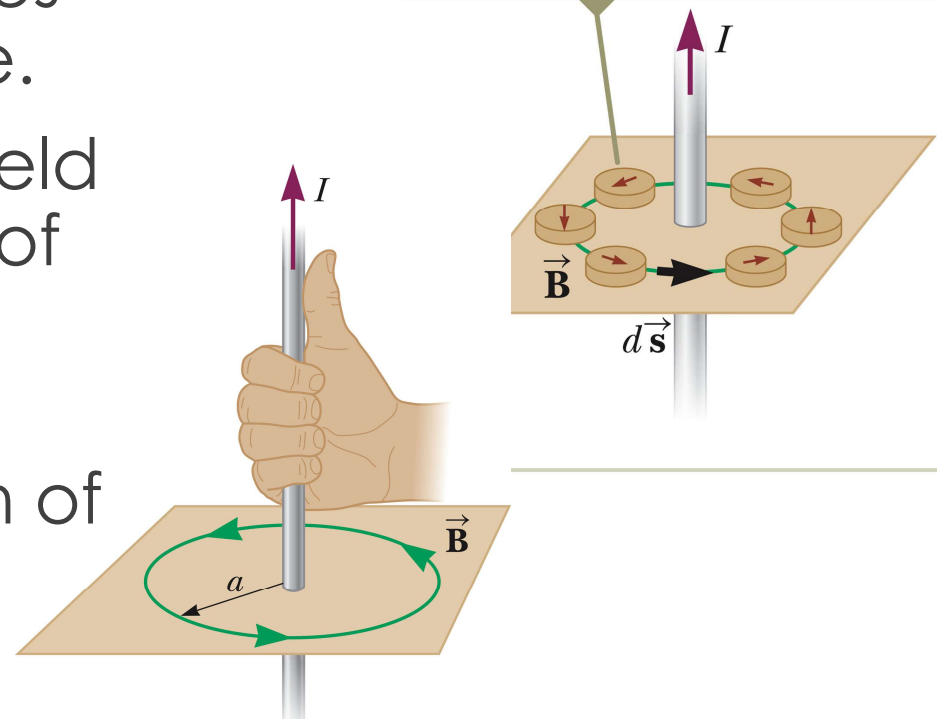
$$\mathbf{B} = \frac{\mu_0 I}{4\pi a} [\cos \theta_1 - \cos \theta_2] \hat{\mathbf{k}}$$



Magnetic Field for a Long, Straight Conductor: Direction

- ▶ The magnetic field lines are circles concentric with the wire.
- ▶ The field lines lie in planes perpendicular to the wire.
- ▶ The magnitude of the field is constant on any circle of radius a .
- ▶ The right-hand rule for determining the direction of the field is shown.

When the wire carries a strong current, the compass needles deflect in a direction tangent to the circle, which is the direction of the magnetic field created by the current.

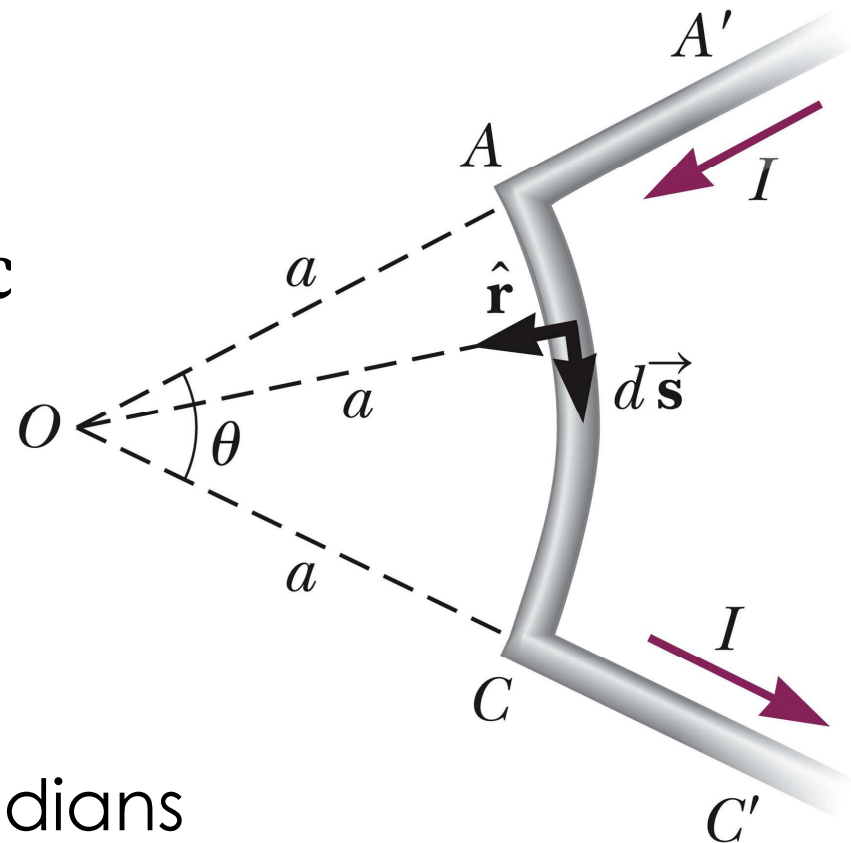


Magnetic Field for a Curved Wire Segment

$$\begin{aligned}\therefore B &= \frac{\mu_0}{4\pi} \int \frac{Idl}{R^2} = \frac{\mu_0 I}{4\pi R^2} \int dl \\ &= \frac{\mu_0 I}{4\pi R^2} \text{ (the length } s \text{ of the arc)} \\ \therefore B &= \frac{\mu_0 I}{4\pi R^2} R\theta = \frac{\mu_0 I}{4\pi R} \theta\end{aligned}$$

$$B = \frac{\mu_0 I}{4\pi R} \theta$$

θ will be in radians



Magnetic Field for a Circular Loop of Wire

► Consider the previous result, with a full circle

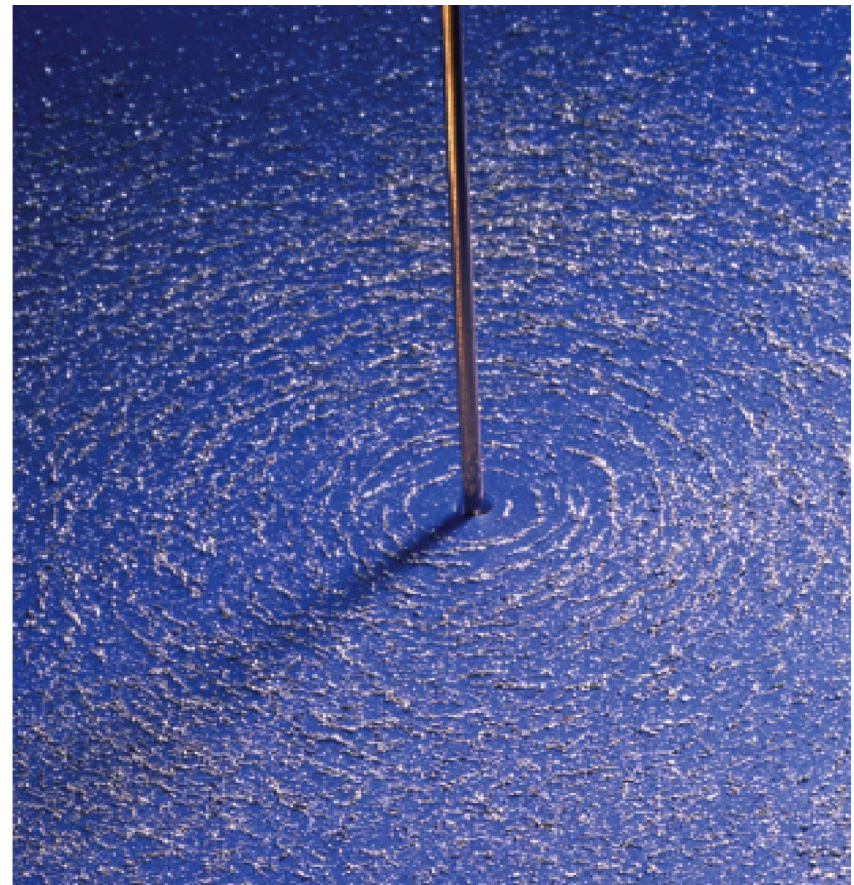
► $\theta = 2\pi$

$$B = \frac{\mu_o I}{4\pi a} \theta = \frac{\mu_o I}{4\pi a} 2\pi = \frac{\mu_o I}{2a}$$

► This is the field at the center of the loop.

Magnetic Field of a Wire

- ▶ The circular magnetic field around the wire is shown by the iron filings.

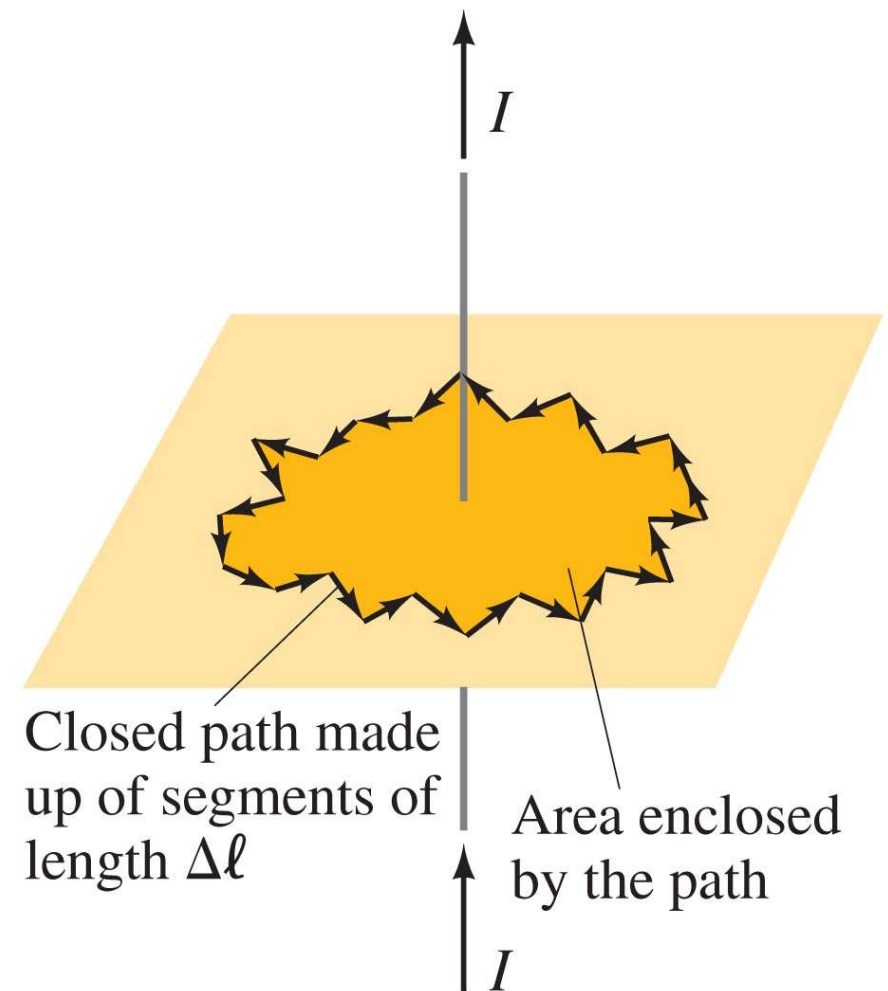


28-4 Ampère's Law

- ▶ Ampère's law relates the magnetic field around a closed loop to the total current flowing through the loop

$$\oint \vec{\mathbf{B}} \cdot d\vec{\ell} = \mu_0 I_{\text{encl.}}$$

- ▶ This integral is taken around the edge of the closed loop.



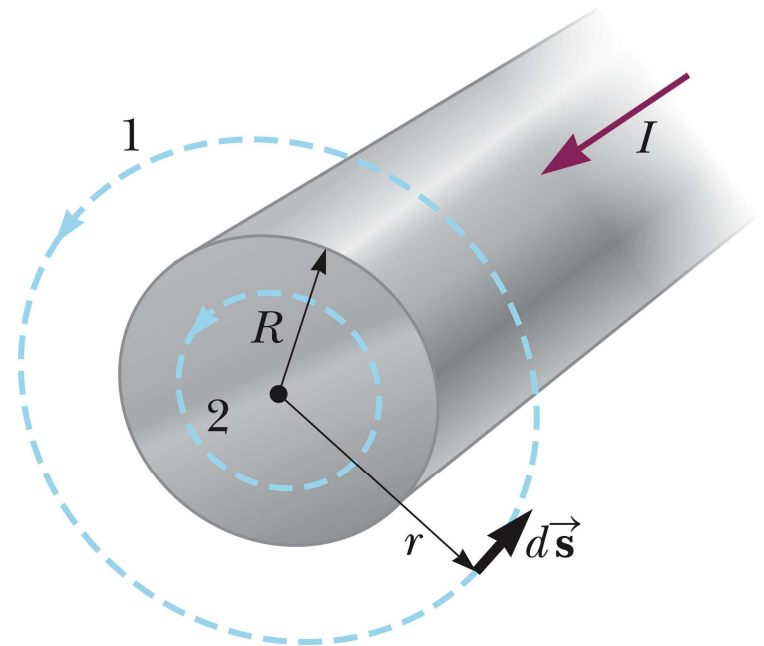
Ampere's Law

- ▶ Ampere's law describes the creation of magnetic fields by all continuous current configurations.
- ▶ Put the thumb of your right hand in the direction of the current through the amperian loop and your fingers curl in the direction you should integrate around the loop.
- ▶ Ampère's law is useful in calculating the magnetic field of a highly symmetric configuration carrying a steady current
- ▶ Choose an integration path that reflects the symmetry (typically, the path is along lines where the field is constant and perpendicular to the field where it is changing)
- ▶ Use the symmetry to determine the direction of the field.
- ▶ Determine the enclosed current.

Field Due to a Long Straight Wire – From Ampere's Law

- ▶ Calculate the magnetic field at a distance r from the center of a wire carrying a steady current I .
- ▶ Since the wire has a high degree of symmetry, the problem can be categorized as a Ampère's Law problem.

$$\begin{aligned}\mu_0 I &= \oint \vec{B} \cdot d\vec{\ell} \\ &= \oint B \, d\ell = B \oint d\ell = B(2\pi r).\end{aligned}$$



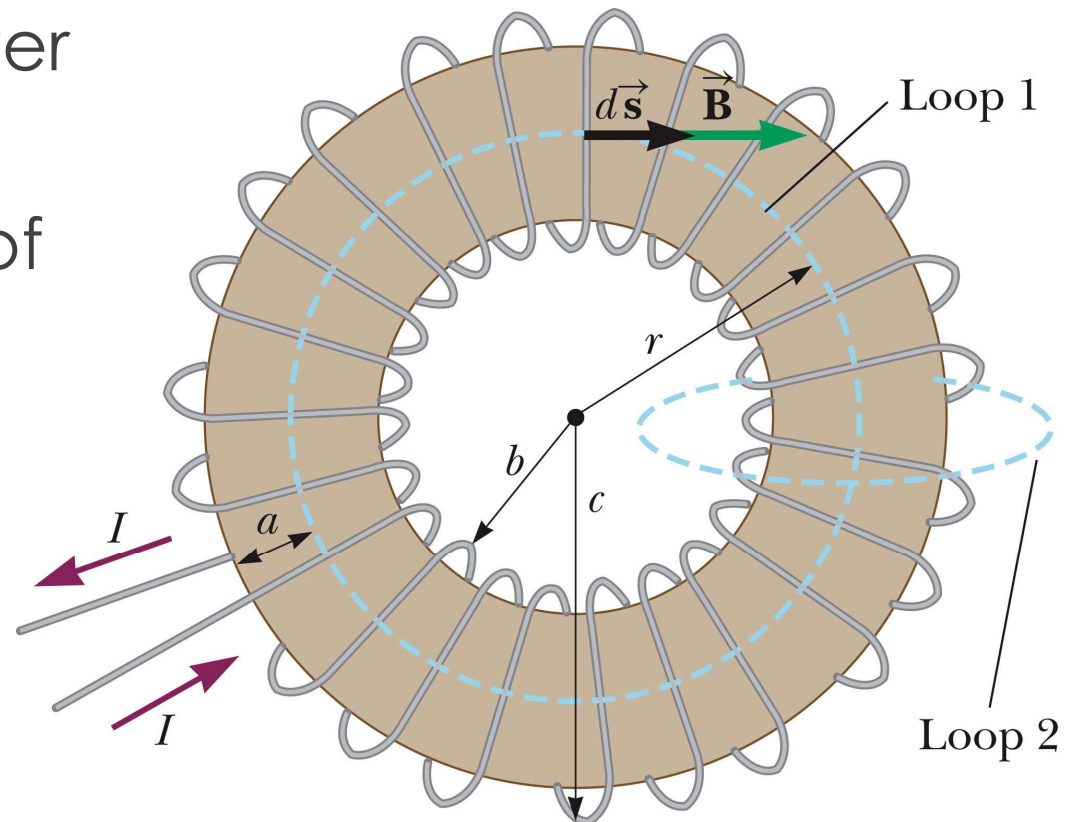
$$B = \frac{\mu_0 I_0}{2\pi r}$$

Magnetic Field of a Toroid

- Find the field at a point at distance r from the center of the toroid.
- The toroid has N turns of wire.

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = B(2\pi r) = \mu_0 N I$$

$$B = \frac{\mu_0 N I}{2\pi r}$$

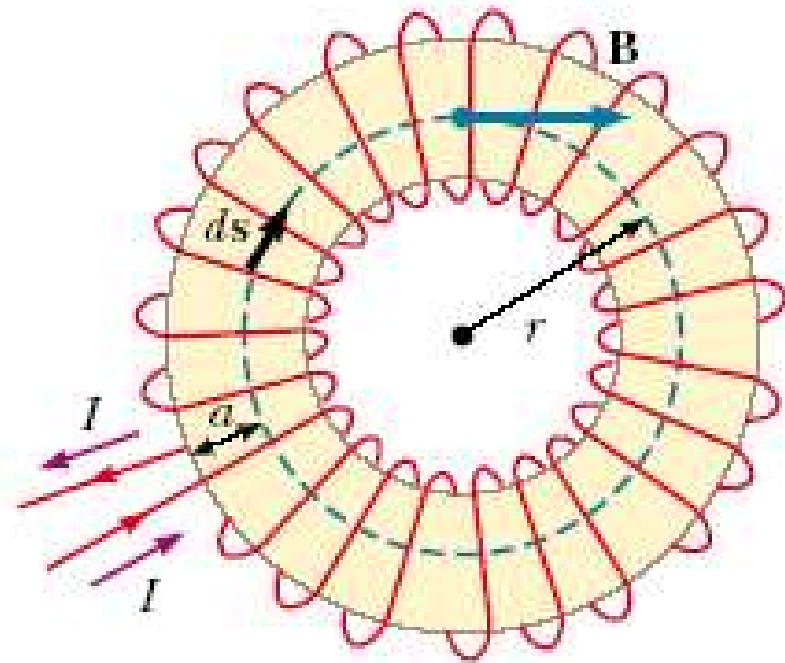


Magnetic Field of a Toroid

► $B = 0$ for

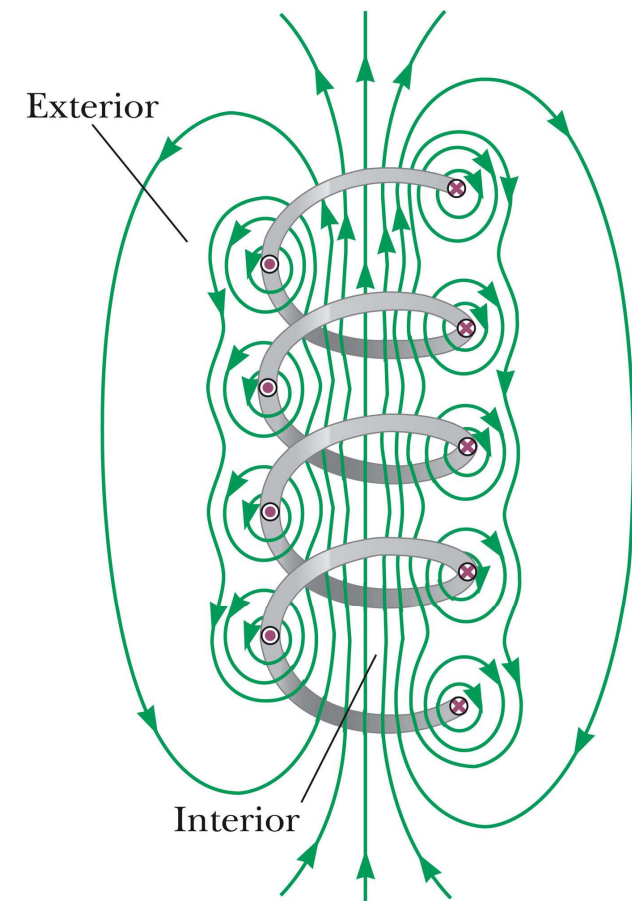
○ $r < R_1$

○ $r > R_2$



Magnetic Field of a Solenoid

- ▶ A solenoid is a long wire wound in the form of a helix.
- ▶ A reasonably uniform magnetic field can be produced in the space surrounded by the turns of the wire.
 - ▶ The interior of the solenoid



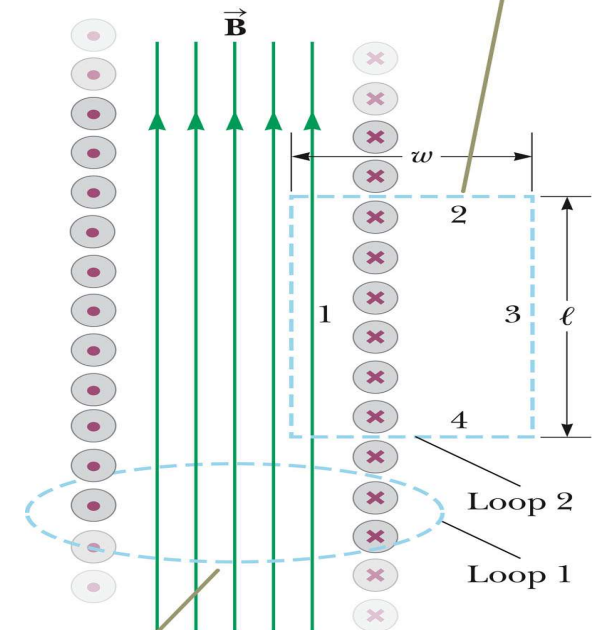
Magnetic Field of a Solenoid

- ▶ The field lines in the interior are
 - ▶ Nearly parallel to each other
 - ▶ Uniformly distributed
 - ▶ Close together
- ▶ This indicates the field is strong and almost uniform.

Infinite Solenoid – Characteristics

- ▶ An ideal solenoid is approached when:
 - ▶ The turns are closely spaced.
 - ▶ The length is much greater than the radius of the turns.

Ampère's law applied to the rectangular dashed path can be used to calculate the magnitude of the interior field.



Ampère's law applied to the circular path whose plane is perpendicular to the page can be used to show that there is a weak field outside the solenoid.

Magnetic Field of a Solenoid

$$\oint \mathbf{B} \cdot d\mathbf{s} = \int_{\text{path1}} \mathbf{B} \cdot d\mathbf{s} + \int_{\text{path2}} \mathbf{B} \cdot d\mathbf{s} + \int_{\text{path3}} \mathbf{B} \cdot d\mathbf{s} + \int_{\text{path4}} \mathbf{B} \cdot d\mathbf{s}$$

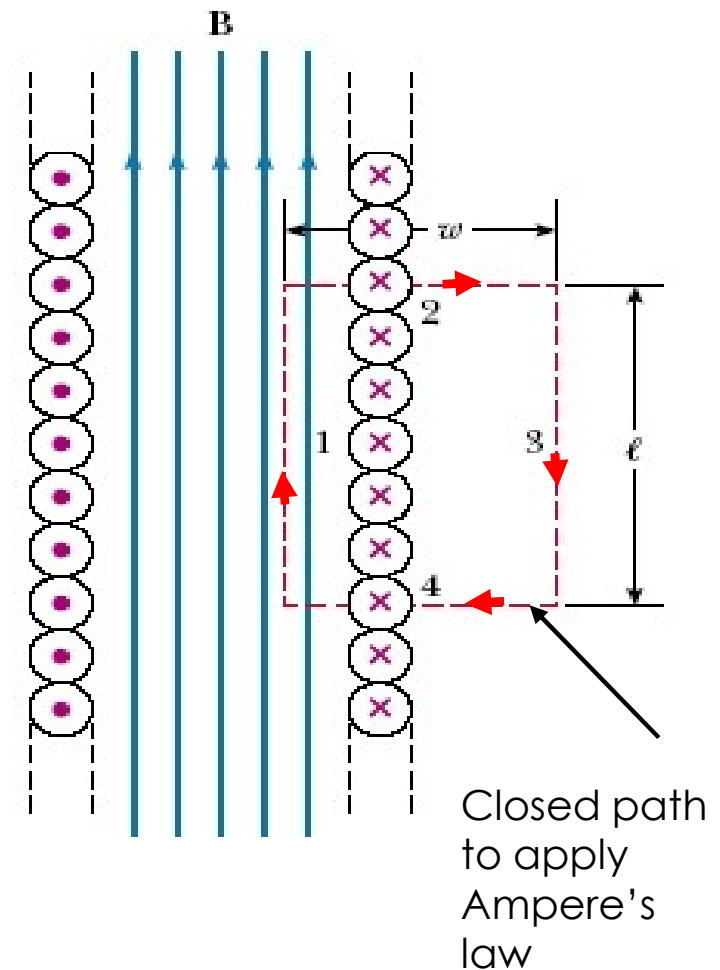
$$\int_{\text{path 1}} \mathbf{B} \cdot d\mathbf{s} = \int_{\text{path 1}} B \cos 0^\circ ds = B \int_{\text{path 1}} ds = Bl$$

$$\int_{\text{path 2}} \mathbf{B} \cdot d\mathbf{s} = 0 = \int_{\text{path3}} \mathbf{B} \cdot d\mathbf{s} = \int_{\text{path4}} \mathbf{B} \cdot d\mathbf{s}$$

$$\therefore \oint \mathbf{B} \cdot d\mathbf{s} = Bl \quad \text{Applying Ampere's law}$$

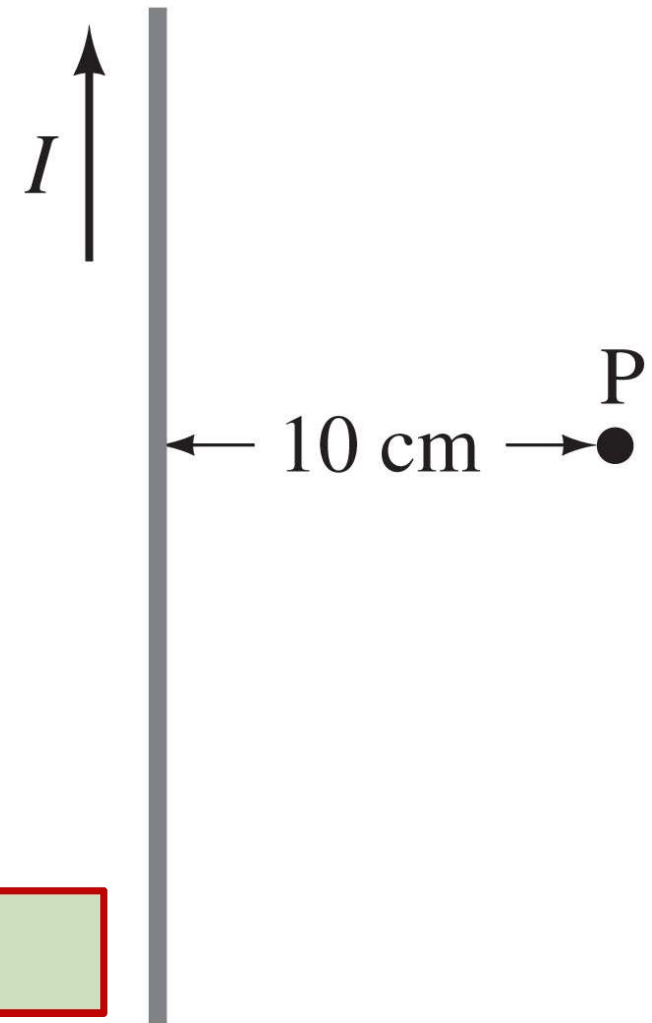
$$\therefore Bl = \mu_0 (nI) \quad \text{where } n \text{ is the number of turns per unit length.}$$

$$B = \mu_0 \frac{N}{\ell} I = \mu_0 n I$$



Example 1:

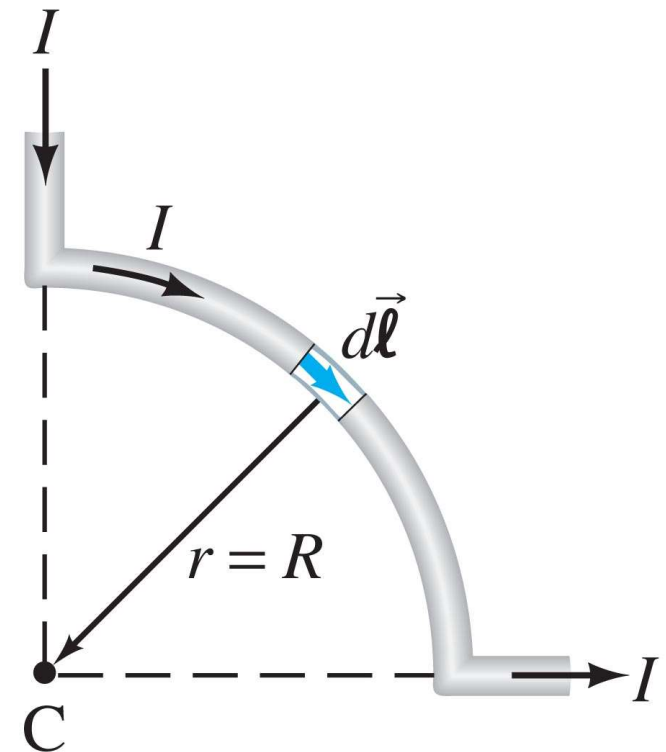
- ▶ An electric wire in the wall of a building carries a dc current of 25 A vertically upward. What is the magnetic field due to this current at a point P 10 cm due north of the wire?



$$\text{Solution : } B = \mu_0 I / 2\pi r = 5.0 \times 10^{-5} \text{ T}$$

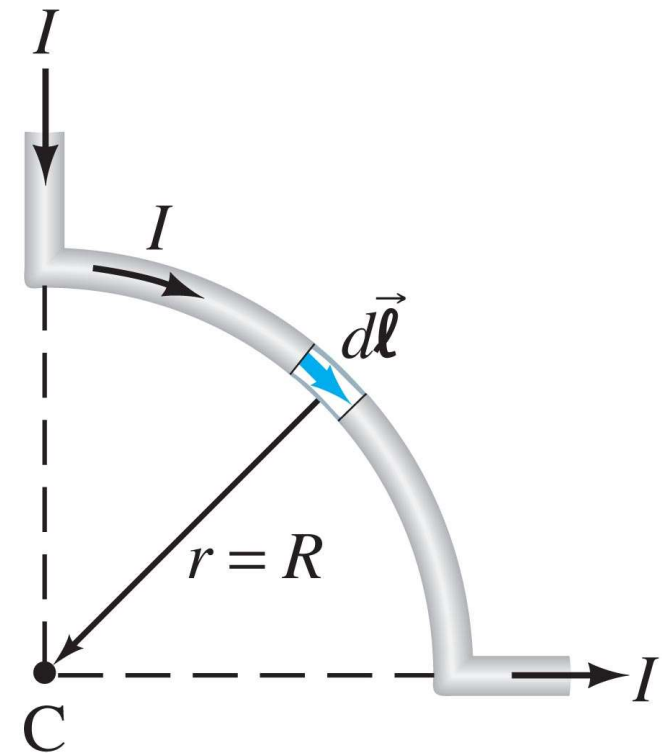
Example 2:

- ▶ **B** due to a wire segment.
- ▶ One quarter of a circular loop of wire carries a current I . The current I enters and leaves on straight segments of wire, as shown; the straight wires are along the radial direction from the center C of the circular portion. Find the magnetic field at point C .



Solution

- ▶ The straight segments of wire produce no field at C, as the current is parallel to r . On the curved segment, all points are the same distance (R) from C; the integral is therefore only over $d\ell$, and the result is
- ▶ $B = \mu_0 I / 8R$.



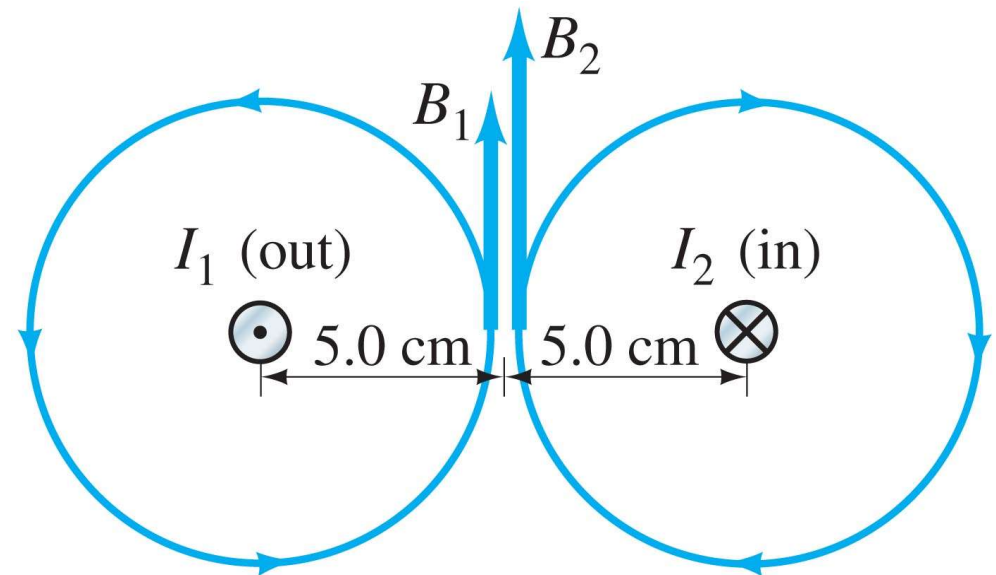
Example 3:

- ▶ A thin 10-cm-long solenoid used for fast electromechanical switching has a total of 400 turns of wire and carries a current of 2.0 A. Calculate the field inside near the center.

Solution: $B = 1.0 \times 10^{-2} \text{ T}$.

Example 4:

- Two parallel straight wires 10.0 cm apart carry currents in opposite directions. Current $I_1 = 5.0$ A is out of the page, and $I_2 = 7.0$ A is into the page. Determine the magnitude and direction of the magnetic field halfway between the two wires.



Solution

- ▶ Wire 1 carrying current I_1 out towards us, and wire 2 carrying current I_2 into the page, produce magnetic fields whose lines are circles around their respective wires.
- ▶ Solution: As the figure shows, the two fields are in the same direction midway between the wires. Therefore, the total field is the sum of the two, and points upward: $B_1 = 2.0 \times 10^{-5} \text{ T}$; $B_2 = 2.8 \times 10^{-5} \text{ T}$; so $B = 4.8 \times 10^{-5} \text{ T}$.

